

SyncLimbs: Semi-automatic Supernumerary Limbs Control System through Motion Synchronization for Enhanced Agency

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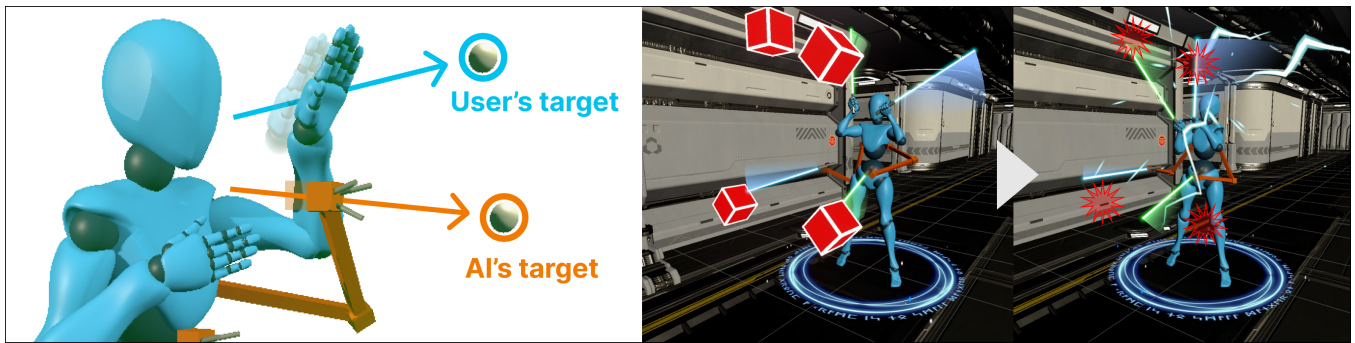


Figure 1: SyncLimbs system overview and demonstration scenario. Left: Four-armed avatar controlled by our system, demonstrating synchronized motion coordination between original limbs and AI-controlled supernumerary limbs. Middle and Right: Demonstration sequence showing synchronized target rhythm task where the user slashes two targets with sabers in the original limbs while AI agents simultaneously control supernumerary limbs to slash remaining targets.

Abstract

Body augmentation through supernumerary limbs faces a control paradox: direct manipulation enhances the sense of agency but increases cognitive load, while automation reduces load but compromises the sense of agency. We propose “SyncLimbs,” a semi-autonomous control system that resolves this trade-off by synchronizing the autonomous motion of additional limbs with user movements. In our approach, supernumerary limbs move so that their task progress is equal to that of the user’s hand, enabling users to perceive causal relationships between their natural movements and the actions of the supernumerary limbs. This probably fosters an enhanced sense of agency without excessive cognitive load.

CCS Concepts

• **Human-centered computing** → **Virtual reality**; • **Computer systems organization** → **Robotic control**.

Keywords

Supernumerary Limb, Semi-automatic Control, Sense of Agency, Virtual Reality

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1 Introduction

Body augmentation technologies represent a transformative approach to enhancing human cognitive and physical capabilities beyond biological limitations. Among various augmentation techniques, supernumerary limbs – additional appendages not naturally present in the human body – constitute a promising method for enhancing capabilities. Supernumerary limbs enable users to perform concurrent multitasking operations [Penaloza and Nishio 2018] and execute tasks at accelerated speeds [Laha et al. 2016], thereby significantly improving overall productivity compared to conventional human-machine interaction paradigms.

Current supernumerary limb control methodologies encompass two primary, yet fundamentally limited, approaches. The first approach utilizes residual degrees of freedom from existing body

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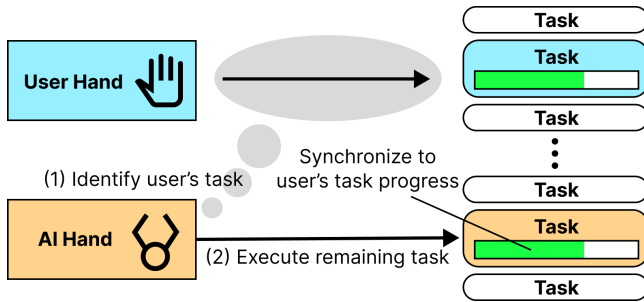


Figure 2: SyncLimbs system workflow. AI hands (1) identify which tasks the user hands are executing. Then they (2) select and execute their task while synchronizing their task progress to the user’s task.

parts, including head movements [Laha et al. 2016], legs [Sasaki et al. 2017], or neural signals through brain-computer interfaces [Penaloza and Nishio 2018]. The second approach delegates control to external agents, such as additional human operators [Zhou et al. 2025] or motion generation AI [Parietti and Asada 2014].

These existing control approaches create a fundamental trade-off between cognitive and physical load that limits their practical effectiveness and the sense of agency (“the subjective feeling of controlling one’s actions, and through them, external events” [Wen and Imamizu 2022]). While the former methods facilitate the emergence of the sense of agency, they also impose substantial cognitive load through simultaneous management of multiple limb-specific tasks and elevated physical load for controlling them. Conversely, the latter methods substantially reduce user workload but compromise user agency and intentional control, thereby inhibiting agency. Parallel Ping-Pong [Takada et al. 2022] or MultiSoma [Miura et al. 2022] are other methods that successfully control multiple robots or avatars by duplicating the user’s movement to them. However, to leverage the advantages of additional limbs, such as for multitasking, these limbs should simultaneously perform different types of movements.

We propose “SyncLimbs,” a semi-autonomous control system that synchronizes motion between supernumerary limbs and user movements (Figure 1 left). In this system, the user’s hands and the AI’s hands perform different kinds of movements simultaneously, while the AI’s hands ensure that their task progress is synchronized with the user’s hand task. Our approach is based on the hypothesis that synchronizing autonomous movements of supernumerary limbs with user movements enhances the sense of agency without imposing excessive physical and cognitive load. This hypothesis is based on prior research suggesting that synchronized movement with other people generates a sense of agency [Hao et al. 2020].

2 Proposed Method

The SyncLimbs system architecture comprises user hands and AI hands, enabling the coordinated execution of tasks (See Figure 2). User hands correspond to the avatar’s original hands and receive direct input from user movements, while AI hands semi-automatically control the endpoints of supernumerary limbs.

Task representation within the system utilizes atomic action units (e.g., moving to the specific destination), enabling precise

progress tracking and control over synchronization. They need definitions of a start, a goal, and a progress rate value between them, ranging from 0 to 1, that can be calculated individually for each hand. AI hands calculate the progress of the user hand for each task and identify the most advanced one as the user’s task. Then they select the most advanced tasks for themselves to execute.

AI hands execute tasks through adaptive motion generation algorithms that maintain synchronization with the user hand’s progress. They dynamically switch task-specific motion planning strategies that we implement. For spatial navigation tasks that require movement to designated positions, for example, the system employs linear-interpolation-based motion generation. AI hand moves to the interpolated position between its destination and default position, where the parameter is the task progress of the user hand.

3 Demonstration at SIGGRAPH Asia

Our demonstration will showcase our system through multitasking experiences within a controlled virtual environment. Participants will engage in object-reaching tasks with numerous dynamically appearing targets, as well as a target rhythm task where four arms must slice moving objects in real-time (See Figure 1 middle and left). These experiences enable users to discover their enhanced capabilities to accomplish seemingly impossible tasks and the practical advantages of an augmented sense of agency. Through these demonstrations, they will gain a firsthand understanding of how semi-autonomous control can enhance human task performance while maintaining intuitiveness.

Acknowledgments

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